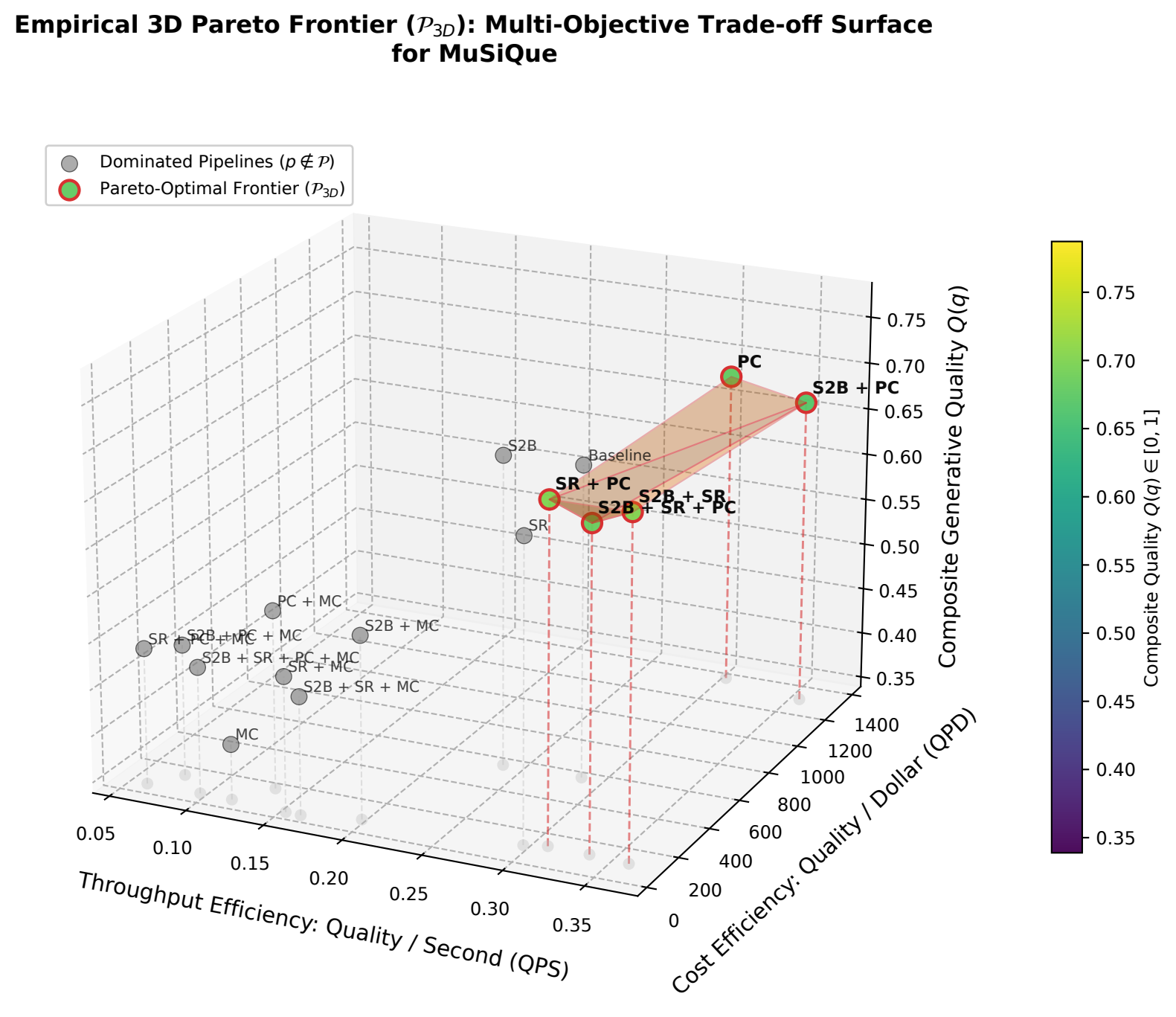
Empirical 3D Pareto Frontier ($\mathcal{P}_{3D}$): Multi-Objective Trade-off Surface for MuSiQue
Dominated Pipelines ($p \notin \mathcal{P}$)
Pareto-Optimal Frontier ($\mathcal{P}_{3D}$)
PC
S2B + PC
S2B
Baseline
SR + PC
S2B + SR
S2B + SR + PC
SR
PC + MC
S2B + MC
SR + PC + MC
S2B + PC + MC
S2B + SR + PC + MC
SR + MC
S2B + SR + MC
MC
Throughput Efficiency: Quality / Second (QPS)
Cost Efficiency: Quality / Dollar (QPD)
Composite Generative Quality $Q(q)$
Composite Quality $Q(q) \in [0, 1]$
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0
200
400
600
800
1000
1200
1400
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
0.75